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FLEXIBLE DISPLAY PANEL AND DISPLAY APPARATUS INCLUDING THE FLEXIBLE DISPLAY PANEL

Abstract

A flexible display panel including: a flexible panel including a first region including a display region and oriented on a first plane and a second region including a non-display region and oriented on a second plane different from the first plane; and a flexible encapsulation member disposed on the flexible panel so as to encapsulate at least the display region.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/498,022, filed Oct. 30, 2023, which is a continuation of U.S. patent application Ser. No. 17/521,645, filed Nov. 8, 2021, now U.S. Pat. No. 11,803,210, which is a continuation of U.S. patent application Ser. No. 16/701,110, filed Dec. 2, 2019, now U.S. Pat. No. 11,169,570, which is a continuation of U.S. patent application Ser. No. 16/114,120, filed Aug. 27, 2018, now U.S. Pat. No. 10,496,133, which is a continuation of U.S. patent application Ser. No. 13/535,298, filed Jun. 27, 2012, now U.S. Pat. No. 10,061,356, which claims priority to and the benefit of Korean Patent Application Nos. 10-2011-0065143, filed Jun. 30, 2011, and 10-2012-0063871, filed Jun. 14, 2012, the entire content of all of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] One or more embodiments of the present invention relate to a flexible display panel and a display apparatus including the flexible display panel.

2. Description of the Related Art

[0003] Display apparatuses refer to apparatuses that display an image signal. Such

[0004] display apparatuses include all apparatuses, such as TVs, computer monitors, personal digital assistants (PDAs) and smart devices that are increasingly in demand, which display an image signal input by an external device.

[0005] A flat panel display module with high image quality, such as an organic light-emitting display panel, a liquid crystal display (LCD) panel, a plasma display panel (PDP), or the like, is used in display apparatuses.

[0006] The flat panel display module includes a display region in which an image is displayed, and a non-display region in which various circuits and wirings for supplying an image signal to the display region are disposed. In this regard, the non-display region is outside the display region on the same plane as that of the display region.

[0007] As demand for high image quality and various applications of display apparatuses is recently increasing, the number of circuits and wirings disposed in the non-display region of the flat panel display module is also increasing. Thus, the area of the non-display region has increased, which causes a reduction in the percentage of the display region that is recognized by a user.

SUMMARY OF THE INVENTION

[0008] One or more embodiments of the present invention provide a flexible display panel that significantly decreases the ratio of non-display regions to a display region recognized by a user, and a display apparatus including the flexible display panel.

[0009] According to an aspect of the present invention, there is provided a flexible

[0010] display panel including: a flexible panel including a first region including a display region and oriented on a first plane and a second region including a non-display region and oriented on a second plane different from the first plane; and a flexible encapsulation member disposed on the flexible panel so as to encapsulate at least the display region.

[0011] The display region may be configured to be viewed from a first direction in

[0012] which the first plane is oriented, and the non-display region may be configured not to be viewed from the first direction.

[0013] An area ratio of the non-display region to the display region visible from a first direction in which the first plane is oriented, may be smaller than an area ratio of the non-display region to the display region visible from the first direction than if the non-display region was oriented in the first direction.

[0014] A border region having a curvature radius may be between the first region and the second region.

[0015] The curvature radius may have a substantially constant value.

[0016] The curvature radius may be between about 0.01 mm and about 10 mm.

[0017] The curvature radius may be between about 0.01 mm and about 1 mm.

[0018] A curved surface of the border region may be arc-shaped.

[0019] The display region may extend into the border region.

[0020] The non-display region may extend into border region.

[0021] Each of the display region and the non-display region may extend into the border region.

[0022] The first region may be substantially planar.

[0023] The second region may be substantially planar.

[0024] The flexible substrate may further include a third region extending from an edge of the second region and extending in a direction towards the first region.

[0025] The non-display region may extend from the second region into the third region.

[0026] A pad unit may be located in the third region.

[0027] The flexible display panel may further include a barrier layer located between the flexible substrate and at least one of the first region and the second region.

[0028] The flexible display panel may further include a first protective film located on one side of the flexible panel and a second protective film located on one side of the flexible encapsulation member.

[0029] The first protective film and the second protective film may be located so as to cover the display region and the non-display region of the flexible panel.

[0030] The first protective film and the second protective film may be located to be symmetrical to each other.

[0031] At least one of the first protective film and the second protective film may include a polarization film.

[0032] The first protective film may cover an edge of the flexible panel, and the second protective film may cover an edge of the flexible encapsulation member.

[0033] The flexible encapsulation member may include at least one of an inorganic insulating layer and an organic insulating layer.

[0034] The flexible display panel may further include a touch panel located on at least one side of the flexible substrate and the flexible encapsulation member and on which an image of the display region is displayed.

[0035] The touch panel may include a flexible film that generally corresponds to a shape of the flexible display panel.

[0036] The touch panel may include a touch region located on the flexible film and [0037] generally corresponding to the display region, and a wiring region outside the touch region. [0038] According to another aspect of the present invention, there is provided a display apparatus including a flexible display panel including a flexible panel including a first region including a display region and oriented on a first plane and a second region including a non-display region and oriented on a second plane different from the first plane, a flexible encapsulation member disposed on the flexible panel so as to encapsulate at least the display region, and a support unit for maintaining a shape of the flexible display panel.

[0039] The support unit may be disposed on the flexible encapsulation member so as to correspond to a shape of the flexible display panel.

[0040] The display apparatus may further include a first protective film located on one side of the flexible panel and a second protective film located on one side of the flexible encapsulation member, wherein the first protective film and the second protective film are located so as to cover the display region and the non-display region of the flexible panel.

[0041] The display apparatus may further include a flexible touch panel located on at least one side of the flexible panel and the flexible encapsulation member and on which an image of the display region is displayed, so as to correspond to a shape of the flexible panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0042] The above and other features and advantages of the present invention will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings in which:

[0043] FIG. 1 is a schematic perspective view of a flexible display panel according to an embodiment of the present invention;

[0044] FIG. 2 is a cross-sectional view of the flexible display panel illustrated in FIG. 1, taken along a line II-II of FIG. 1;

[0045] FIG. 3 is a schematic perspective view of a flat display panel according to a comparative example of FIG. 1;

[0046] FIG. 4 is a cross-sectional view of the flat display panel illustrated in FIG. 3, taken along a line IV-IV of FIG. 3;

[0047] FIG. 5 is a schematic plan view of an example of a flexible display panel before non-display regions of FIG. 1 are folded;

[0048] FIG. 6 is a schematic cross-sectional view of FIG. 5;

[0049] FIG. 7 is a cross-sectional view of an example of display regions of FIG. 5;

[0050] FIG. 8 is a schematic cross-sectional view of a flat display panel before or after being bent according to a comparative example of FIG. 1;

[0051] FIG. 9 illustrates the relationship between an inner curvature radius of a portion in which a display region and a non-display region are folded and widths of the non-display region that are recognized by a user;

[0052] FIGS. 10 and 11 are cross-sectional views of examples of border regions;

[0053] FIG. 12 is a schematic perspective view of a flexible display panel according to another embodiment of the present invention;

[0054] FIG. 13 is a cross-sectional view of the flexible display panel illustrated in FIG. 12, taken along a line XIII-XIII of FIG. 12;

[0055] FIG. 14 is a schematic plan view of an example of a flexible display panel before non-display regions of FIG. 12 are folded;

[0056] FIG. 15 is a schematic perspective view of a flexible display panel according to a modified

example of FIG. 12;

[0057] FIG. 16 is a cross-sectional view of the flexible display panel illustrated in FIG. 14, taken along a line XVI-XVI of FIG. 14;

[0058] FIG. 17 is a schematic perspective view of a flexible display panel according to another embodiment of the present invention;

[0059] FIG. 18 is a cross-sectional view of the flexible display panel illustrated in FIG. 17, taken along a line XVIII-XVIII of FIG. 17;

[0060] FIG. 19 is a schematic plan view of an example of a flexible display panel before non-display regions of FIG. 17 are folded;

[0061] FIG. 20 is a schematic perspective view of a flexible display panel according to a modified example of FIG. 17;

[0062] FIG. 21 is a cross-sectional view of the flexible display panel illustrated in FIG. 19, taken along a line XXI-XXI of FIG. 20;

[0063] FIG. 22 is a cross-sectional view of a flexible display panel-integrated touch panel according to another embodiment of the present invention;

[0064] FIG. 23 is a cross-sectional view of a flexible display panel-integrated touch panel before non-display regions of the flexible display panel-integrated touch panel illustrated in FIG. 22 are folded;

[0065] FIG. 24 is an exploded perspective view of the flexible display panel-integrated touch panel illustrated in FIG. 23; and

[0066] FIGS. 25 through 27 are schematic cross-sectional views of examples in which a flexible display panel is combined with a variety of support units.

DETAILED DESCRIPTION OF THE INVENTION

[0067] The present invention will now be described more fully with reference to the accompanying drawings in which exemplary embodiments of the invention are shown.

[0068] FIG. 1 is a schematic perspective view of a flexible display panel 100 according to an embodiment of the present invention, and FIG. 2 is a cross-sectional view of the flexible display panel 100 illustrated in FIG. 1, taken along a line II-II of FIG. 1.

[0069] Referring to FIGS. 1 and 2, the flexible display panel 100 includes a display region D oriented on a first plane (x-y plane), and non-display regions N1, N2, N3 and N4 oriented on a second plane (y-z plane) different from the first plane (x-y plane) and outside the display region D. The non-display regions N1 and N3 that face each other across the display region D among the non-display regions N1, N2, N3 and N4 are folded in a direction-z opposite to a direction z in which the display region D is displayed.

[0070] Since the non-display regions N1 and N3 are folded in the direction-z opposite to the direction z in which the display region D is displayed, when the user sees the flexible display panel 100 from the front (z-direction) of the display region D, the user recognizes a width $W_{sub.1}$ at which the non-display regions N1 and N3 are folded as a border that surrounds right and left sides of the display region D.

[0071] Referring to FIG. 2, since the width $W_{sub.1}$ at which the non-display regions N1 and N3 are folded becomes much smaller than an original width $W_{sub.0}$ before the non-display regions N1 and N3 are folded, the user recognizes that the border of the display region D formed by the non-display regions N1 and N3 is significantly decreased in size. When there is no change in the area of the display region D before and after the non-display regions N1 and N3 are folded, respectively, the area ratio of the non-display regions N1 and N3 to the display region D recognized by the user is decreased, and thus, the effects of relatively enlarging the display region D may be shown. Since a width of the display region D may be enlarged by the decreased width ($W_{sub.0}-W_{sub.1}$) of the non-display regions N1 and N3, when the flexible display panel 100 is applied to a display apparatus having the same specification as that of the flexible display panel 100, the flexible display panel 100 may implement an enlarged image screen compared to other display panels.

[0072] The folded width $W_{sub.1}$ of the non-display regions N1 and N3 may be determined by the sum of an inner curvature radius R of a portion in which the display region D and the non-display regions N1 and N3 are folded and a thickness T of the flexible display panel 100. Although described below, since the inner curvature radius R of the portion in which the display region D and the non-display regions N1 and N3 are folded and the thickness T of the flexible display panel 100 may be significantly decreased, the user may recognize the display region D formed by the non-display regions N1 and N3 as being almost borderless.

[0073] FIG. 3 is a schematic perspective view of a flat display panel 10 in which non-display regions N1, N2, N3 and N4 are not folded as a comparative example of FIG. 1, and FIG. 4 is a cross-sectional view of the flat display panel 10 illustrated in FIG. 3, taken along a line IV-IV of FIG. 3.

[0074] Referring to FIGS. 3 and 4, the flat display panel 10 in which the non-display regions N1, N2, and N3 and N4 are not folded includes a display region D and the non-display regions N1, N2, N3, N4 outside the display region D.

[0075] Since the non-display regions N1 and N3 are not folded, when the user sees the flat display panel 10 from the front (z-direction) of the display region D, the user recognizes the original width $W_{sub.0}$ of the non-display regions N1 and N3 as a border that surrounds the right and left sides of the display region D. Thus, a thicker border of the display region D formed by the non-display regions N1 and N2 is recognized by the user, compared to the flexible display panel 100 of FIG. 1 described above.

[0076] When the area of the display region D and the non-display regions N1 and N3 of the flexible display panel 100 of FIG. 1 is the same as the area of the display region D and the non-display regions N1 and N3 of the flat display panel 10 of FIG. 3, the area ratio of the non-display regions N1 and N3 to the display region D of the flexible display panel 100 of FIG. 1 recognized by the user is decreased compared to that of the flat display panel 10 of the comparative example so that the effects of enlarging a display screen may be shown.

[0077] In the case of the case of the n the case of the flat display panel 10 according to the comparative example, in order to reduce the area of the non-display regions N1 and N3, the number of various circuits or wirings disposed in the non-display regions N1 and N3 has to be decreased, or line widths of circuits or wirings have to be decreased. However, since demand for high image quality and various applications of a display apparatus is increasing, there is a limitation in reducing the number of circuits or wirings. In addition, there is a process difficulty in reducing line widths of the circuits or wirings. However, in the flexible display panel 100 of FIG. 1, the area of the non-display regions N1 and N3 recognized by the user may be decreased without reducing the number of circuits or wirings disposed in the non-display regions N1 and N3 or without reducing line widths of the circuits or wirings.

[0078] FIG. 5 is a schematic cross-sectional plan view of an example of a flexible display panel 100-B before the non-display regions N1 and N3 of FIG. 1 are folded, and FIG. 6 is a schematic cross-sectional view of FIG. 5, and FIG. 7 is a cross-sectional view of an example of a display region D of FIG. 5.

[0079] Referring to FIGS. 5 through 7, the flexible display panel 100-B before the non-display regions N1 and N3 are folded includes a flexible substrate 111, a barrier layer 112 located on the flexible substrate 111, the display region D and non-display regions N1, N2, N3 and N4 disposed on the barrier layer 112, a flexible encapsulation member 113, a first protective film 114 located on one side of the flexible substrate 111, and a second protective film 115 located on one side of the flexible encapsulation member 113.

[0080] The flexible substrate 111 may be formed of a plastic material having excellent heat resistance and durability, such as polyethylene ether phthalate, polyethylene naphthalate, polycarbonate, polyarylate, polyetherimide, polyethersulfone, polyimide, or the like. However, the present invention is not limited thereto, and the flexible substrate 111 may be formed of various

flexible materials.

[0081] The barrier layer **112** may be located on the flexible substrate **111**. The barrier layer **112** may be formed of at least one of an inorganic layer and an organic layer. The barrier layer **112** prevents unnecessary components from transmitting into the flexible substrate **111** and from permeating into the display region D.

[0082] The display region D and the non-display regions N1, N2, N3 and N4 outside the display region D may be disposed on the barrier layer **112**.

[0083] An image signal is displayed in the display region D, and various display devices (not shown), such as organic light-emitting devices, liquid crystal display (LCD) devices, electrophoresis devices, and the like may be disposed in the display region D. In the current embodiment, at least one organic light-emitting device **120** will be described. Various devices, such as thin film transistors (TFTs) (not shown) and capacitors (not shown) for driving display devices and the like may also be disposed in the display region D.

[0084] The organic light-emitting device **120** is disposed in the display region D. The organic light-emitting device **120** includes a pixel electrode **121**, an opposite electrode **123** that is a common layer, and an organic emission layer **122** interposed between the pixel electrode **121** and the opposite electrode **123**. The pixel electrode **121** is electrically connected to a driving TFT **TFT1** disposed on the flexible substrate **111**. Although not shown in FIG. 6, the organic light-emitting device **120** is electrically connected to at least one switching TFT and storage capacitor, thereby emitting light from the display region D.

[0085] Light may be emitted from the organic emission layer **122** towards the flexible substrate **111** or the flexible encapsulation member **113**. In the current embodiment, a top-emission type display device in which an image is displayed towards the flexible encapsulation member **113** is described. However, the present invention is not limited thereto and may be applied to a bottom-emission type display device in which an image is displayed towards the flexible substrate **111**. In the case of the bottom-emission type display device, the non-display regions N1 and N3 are folded in a direction opposite to a direction-z in which an image is displayed, that is, towards the flexible encapsulation member **113** (z-direction).

[0086] Various circuits (not shown) and wirings (not shown) for supplying an image signal to the display device disposed in the display region D may be disposed in the non-display regions N1, N2, N3 and N4.

[0087] Referring to FIG. 5, an electrode power supply line **131** for supplying power to the opposite electrode **123**, and a terminal portion **132** of the electrode power supply line **131** are disposed in the non-display regions N1, N2, N3 and N4. A scan circuit unit **133** for transmitting a driving signal to the display region D and a terminal portion **134** of the scan circuit unit **133** are disposed in the non-display regions N1 and N2. A data circuit unit **135** for transmitting a data signal to the display region D and a terminal portion **136** of the data circuit unit **135** are disposed in the non-display regions N2 and N3. A driving power wiring unit **137** for supplying driving power to the display region D and a terminal portion **138** of the driving power wiring unit **137** are disposed in the non-display region N2. A pad unit P, on which the terminal portions **132**, **134**, **136**, and **138** are disposed, is located in the non-display region N2.

[0088] The various circuits, wirings, and terminal portions illustrated in FIG. 5 are just illustrative of an example of the non-display regions N1, N2, N3 and N4. In other words, obviously, various circuits, wirings, and terminal portions disposed in the non-display regions N1, N2, N3 and N4 may have different configurations from those of FIG. 5.

[0089] The flexible encapsulation member **113** for encapsulating at least the display region D may be disposed in the display region D and the non-display regions N1, N2, N3 and N4. The flexible encapsulation member **113** may be an encapsulation thin layer. The encapsulation thin layer may have a structure including a plurality of inorganic insulating layers or a structure in which an inorganic insulating layer and an organic insulating layer are alternately stacked.

[0090] The first protective film **114** may be located on one side of the flexible substrate **111**. Although not shown, an adhesive may be further injected between the flexible substrate **111** and the first protective film **114**. The second protective film **115** may be located on one side of the flexible encapsulation member **113**. An adhesive may be further injected between the flexible encapsulation member **113** and the second protective film **115**.

[0091] The first protective film **114** and the second protective film **115** may have the same material property and may serve to improve a mechanical strength of the flexible display panel **100**. The first protective film **114** and the second protective film **115** may be formed of flexible material.

[0092] In addition, at least one of the first protective film **114** and the second protective film **115** may have a polarization function. Here, the arrangement of a protective film having a polarization function may be determined based on a direction in which the organic light-emitting device **120** emits light. For example, when the organic light-emitting device **120** is a top-emission type display device which emits light towards the second protective film **115** so as to display an image, the second protective film **115** is configured to have a polarization function. Meanwhile, when the organic light-emitting device **120** is a bottom-emission type display device which emits light towards the first protective film **114** so as to display an image, the first protective film **114** is configured to have a polarization function. In addition, when the organic light-emitting device **120** is a top and bottom emission type display device, both the first protective film **114** and the second protective film **115** may have a polarization function.

[0093] FIG. **8** is a schematic cross-sectional view of a flat display panel **20** before or after being bent according to a comparative example of FIG. **1**.

[0094] When the second protective film **115** has a polarization function, according to the related art, the first protective film **114** is located to cover the flexible substrate **111** fully, as illustrated in FIG. **8**, whereas the second protective film **115** is located to cover only the display region D fully.

[0095] In this state, when the flat display panel **20** is bent, in the non-display regions N1 and N3, a bending moment varies between a portion including the second protective film **115** and a portion that does not include the second protective film **115**. Thus, cracks may occur in a region C in which metal layers, such as various circuits and wirings for forming the non-display regions N1 and N3 between the portion including the second protective film **115** and the portion that does not the second protective film **115**, are formed. The cracks may induce abnormal lighting of the flat display panel **20**, which may be a cause of defects.

[0096] In order to prevent this problem, in the present embodiment, the first protective film **114** and the second protective film **115** are located to cover the display region D and the non-display regions N1 and N3 of the flexible display panel **100-B**. The first protective film **114** and the second protective film **115** may be located to be symmetrical to each other. For example, the first protective film **114** may be located so as to cover both ends of the flexible substrate **111**, and the second protective film **115** may be located so as to cover both ends of the flexible encapsulation member **113**.

[0097] Through the above structure, the same bending moment may be applied to a region in which metal layers, such as various circuits and wirings for forming the non-display regions N1 and N3 folded at a border between the non-display regions N1 and N3 and the display region D, so that damage or cracks of the region in which metal layers are formed, may be prevented.

[0098] Since the flexible substrate **111** that constitutes the flexible display panel **100**, is flexible, it is difficult to form the display region D and the non-display regions N1 and N3 directly on the flexible substrate **111**. Thus, although not shown, a method of manufacturing the flexible display panel **100** includes: forming the flexible substrate **111** on a glass substrate (not shown) having rigidity; forming a barrier layer **112** on the flexible substrate **111**; forming the display region D and the non-display regions N1 and N3 on the barrier layer **112**; forming the flexible encapsulation member **113** so as to cover the display region D; attaching the second protective film **115** to the flexible encapsulation member **113** by using a second adhesive (not shown); separating the glass

substrate (not shown) from the flexible substrate **111**; and attaching the first protective film **114** to the flexible substrate **111** by using a first adhesive (not shown).

[0099] The first protective film **114** and the second protective film **115** according to the present embodiment may be applied to other embodiments of the present invention to be described as below.

[0100] When the non-display regions **N1** and **N3** on right and left sides of the display region **D** of the flexible display panel **100-B** of FIG. **5** are folded in a direction-z opposite to a direction (z-direction, see FIG. **1**) in which the display region **D** is displayed, the flexible display panel **100** illustrated in FIGS. **1** and **2** is constituted.

[0101] In the flexible display panel **100**, when the non-display regions **N1** and **N3** are folded, a curved surface may be formed at a border between the display region **D** and the non-display regions **N1** and **N3**.

[0102] Referring back to FIG. **2**, a border region **RB** including a predetermined curved surface is between the display region **D** (first region **RI**) and the non-display region **N3** (second region **RII**). The border region **RB** is formed as a curved surface having the inner curvature radius **R**.

[0103] In the current embodiment, the non-display region **N3** of the second region **RII** extends from the border region **RB**. When the user sees the flexible display panel **100** folded in the front (z-direction) of the display region **D**, the width $W_{sub.1}$ of the non-display region **N3** recognized by the user from the front of the display region **D** is recognized as the sum of the inner curvature radius **R** of the portion in which the display region **D** and the non-display region **N3** are folded and the thickness **T** of the flexible display panel **100**. In the current embodiment, the width $W_{sub.1}$ of the non-display region **N3** recognized by the user is the same as the border region **RB**.

[0104] The thickness **T** of the flexible display panel **100** may be selected in a range of several tens of micrometers (μm) and several hundreds of micrometers (μm). In particular, since, in the case of a small display apparatus, the thickness of a flexible display panel may be selected from a narrower range than the range described above, the width $W_{sub.1}$ of the non-display region recognized by the user may be further decreased. Obviously, the present invention is not limited to the thickness range described above, and a wider thickness range than the thickness range described above may be selected if necessary.

[0105] FIG. **9** illustrates the relationship between the inner curvature radius **R** of the portion in which the display region **D** and the non-display region **N3** are folded and widths $W_{sub.2}$ and $W_{sub.3}$ of the non-display region **N3** recognized by a user.

[0106] Referring to FIG. **9**, when it is assumed that the thickness **T** of the flexible display panel **100** is uniform and the non-display region **N3** is folded so that the border region **RB** has a curvature radius $R_{sub.2}$, the width $W_{sub.2}$ of the non-display region **N3** recognized by the user is the sum of the curvature radius $R_{sub.2}$ and the thickness **T** of the flexible display panel **100** ($W_{sub.2}=R_{sub.2}+T$). When the non-display region **N3** is folded so that the border region **RB** has a larger curvature radius $R_{sub.3}$ than the curvature radius $R_{sub.2}$, the width $W_{sub.3}$ of the non-display region **N3** recognized by the user is the sum of the curvature radius $R_{sub.3}$ and the thickness **T** of the flexible display panel. In other words, as the curvature radius increases ($R_{sub.3}>R_{sub.2}$), the width $W_{sub.3}$ of the non-display region **N3** recognized by the user increases.

[0107] The inner curvature radius of the portion in which the display region **D** and the non-display region **N3** are folded may be between 0.01 mm and 10 mm. When the inner curvature radius of the portion is less than 0.01 mm, due to bending stress, the flexible display panel **100** may not be maintained in a folded state, and when the inner curvature radius of the portion is greater than 10 mm, the width $W_{sub.3}$ of the non-display region **N3** increases. In particular, in the case of a small display apparatus, the range of the curvature radius may be selected as being less than 1 mm so that the width $W_{sub.3}$ of the non-display region **N3** recognized by the user may be further decreased.

[0108] Referring to FIG. **10**, the display region **D** of the first region **RI** extends from the border

region RB. When the user sees the flexible display panel **100** folded in the front of the display region D, a width $W_{sub.4}$ of the non-display region N3 recognized by the user is 0. Thus, in the current embodiment, the user may recognize that the non-display region N3 is almost borderless. [0109] Referring to FIG. 11, both the display region D of the first region RI and the non-display region N3 of the second region RII may be disposed in the border region RB. In this regard, a width $W_{sub.5}$ of the non-display region N3 recognized by the user is greater than the width $W_{sub.4}$ of the non-display region N3 recognized by the user, as illustrated in FIG. 10 and is less than the width $W_{sub.1}$ of the non-display region N3 recognized by the user, as illustrated in FIG. 2. However, even in this case, the width $W_{sub.5}$ of the non-display region N3 recognized by the user is recognized as being smaller than that of the flat display panel **10** of FIGS. 3 and 4 in which the non-display regions N1 and N3 are not folded.

[0110] In the case of the flexible display panel **100** of FIG. 1, the non-display regions N1 and N3 on the right and left sides of the display region D are folded. However, the present invention is not limited thereto. In other words, the spirit of the present invention also applies to a case where any one of the non-display regions N1 and N3 is folded.

[0111] FIG. 12 is a schematic perspective view of a flexible display panel **200** according to another embodiment of the present invention, FIG. 13 is a cross-sectional view of the flexible display panel **200** illustrated in FIG. 12, taken along a line XIII-XIII of FIG. 12, and FIG. 14 is a schematic plan view of an example of a flexible display panel **200-B** before non-display regions N2 and N4 of FIG. 12 are folded.

[0112] Hereinafter, the current embodiment will be described with respect to differences to the embodiments of FIGS. 1 and 12.

[0113] Referring to FIG. 14, in the flexible display panel **200-B** according to the current embodiment, the non-display regions N2 and N4 at upper and lower sides of a display region D is folded in a direction-z opposite to a direction (z-direction) in which the display region D is displayed. As a result, the flexible display panel **200** according to another embodiment of the present invention is as illustrated in FIG. 12.

[0114] Referring to FIGS. 12 and 13, the flexible display panel **200** includes the display region D and non-display regions N1, N2, N3 and N4 outside the display region D. The non-display regions N2 and N4 that face each other across the display region D among the non-display regions N1, N2, N3 and N4 are folded in a direction-z opposite to a direction z in which the display region D is displayed.

[0115] Since the non-display regions N2 and N4 are folded in the direction-z opposite to the direction z in which the display region D is displayed, when the user sees the flexible display panel **200** from a front (z-direction) of the display region D, the user recognizes a width $W_{sub.6}$ at which the non-display regions N2 and N4 are folded as a border that surrounds the upper and lower sides of the display region D.

[0116] As described above, since the width $W_{sub.6}$ at which the non-display regions N2 and N4 are folded becomes much smaller than an original width $W_{sub.0}$ before the non-display regions N2 and N4 are folded, the user recognizes that the border of the display region D formed by the non-display regions N2 and N4 is significantly decreased. In particular, since the area of the non-display region N2 in which a pad unit P is located is large, the non-display region N2 in which the pad unit P is located is folded so that the user recognizes that the border of the display region D formed by the non-display region N2 is significantly decreased.

[0117] FIG. 15 is a schematic perspective view of a flexible display panel **200-1** according to a modified example of FIG. 12, and FIG. 16 is a cross-sectional view of the flexible display panel **200-1** illustrated in FIG. 14, taken along a line XVI-XVI of FIG. 14.

[0118] Referring to FIGS. 15 and 16, a non-display region N22 of the flexible display panel **200** illustrated in FIG. 12 in which the pad unit P of the non-display region N2 is located is folded towards an inside of the flexible display panel **200-1** once again.

[0119] Since an external driver (not shown) has to be installed at the pad unit P or a connector C has to be connected to the pad unit P, when the external driver is installed at or the connector C is electrically connected to the pad unit P in the state of FIGS. 12 and 13, the width $W_{sub.6}$ of the non-display region N2 that is recognized by the user is increased compared to the width $W_{sub.6}$ of the non-display region N2 of the flexible display panel 200 of FIG. 12.

[0120] However, when, as in the current modified example, a region N22 of the non-display region N2 in which the pad unit P is located is folded towards the inside of the flexible display panel 200-1 once again, the external driver or the connector C is disposed inside the flexible display panel 200-1 so that the width $W_{sub.6}$ of the non-display region N2 recognized by the user is not changed.

[0121] Although the pad unit P is located in the folded non-display region N22 in the above embodiment, aspects of the present invention are not limited thereto. That is, various circuits and wirings described above may be disposed in the folded non-display region N22. FIG. 17 is a schematic perspective view of a flexible display panel 300 according to another embodiment of the present invention, FIG. 18 is a cross-sectional view of the flexible display panel 300 illustrated in FIG. 17, taken along a line XVIII-XVIII of FIG. 17, and FIG. 19 is a schematic plan view of an example of a flexible display panel 300-B before non-display regions N1, N2, N3, and N4 of FIG. 17 are folded.

[0122] Hereinafter, the current embodiment will be described with respect to differences to the embodiments of FIGS. 1, 12, and 17.

[0123] Referring to FIG. 19, in the flexible display panel 300-B according to the current embodiment, non-display regions N1, N2, N3 and N4 at upper and lower sides and right and left sides of a display region D are folded in a direction-z opposite to a direction (z-direction) in which the display region D is displayed. As a result, the flexible display panel 300 according to the current embodiment of the present invention is constituted, as illustrated in FIG. 17.

[0124] Referring to FIGS. 17 and 18, the flexible display panel 300 includes the display region D and non-display regions N1, N2, N3 and N4 outside the display region D. The non-display regions N1, N2, N3 and N4 that face one another across the display region D in upward and downward directions and in right and left directions are folded in a direction-z opposite to a direction z in which the display region D is displayed.

[0125] Since the non-display regions N1, N2, N3 and N4 are folded in the direction-z opposite to the direction z in which the display region D is displayed, when the user sees the flexible display panel 300 from a front (z-direction) of the display region D, the user recognizes that a width $W_{sub.7}$ at which the non-display regions N1, N2, N3 and N4 are folded as a border that surrounds the upper and lower sides and right and left sides of the display region D.

[0126] As described above, since the width $W_{sub.7}$ at which the non-display regions N1, N2, N3 and N4 are folded becomes much smaller than an original width $W_{sub.0}$ before the non-display regions N1, N2, N3 and N4 are folded, the user recognizes that the border of the display region D formed by the non-display regions N1, N2, N3 and N4 is significantly decreased. In particular, since the area of the non-display regions N1, N2, N3, N4 in which pad units P1, P2, P3 and P4 are respectively located is large, the non-display regions N1, N2, N3 and N4 in which the pad units P1, P2, P3 and P4 are respectively disposed are folded so that the user may recognize that the border of the display region D formed by the non-display regions is significantly decreased.

[0127] FIG. 20 is a schematic perspective view of a flexible display panel 300-1 according to a modified example of FIG. 17, and FIG. 21 is a cross-sectional view of the flexible display panel 300-1 illustrated in FIG. 19, taken along a line XXI-XXI of FIG. 20.

[0128] Referring to FIGS. 20 and 21, non-display regions N12 and N32 of the flexible display panel 300 illustrated in FIG. 17 in which the pad units P1, P2, P3 and P4 of the non-display regions N1, N2, N3, and N4 are located are folded towards an inside of the flexible display panel 300-1 once again.

[0129] Since an external driver (not shown) has to be installed at each of the pad units P1, P2, P3 and P4 or a connector C has to be electrically connected to each of the pad units P1, P2, P3 and P4, when the external driver is installed at or the connector C is electrically connected to each of the pad units P1, P2, P3 and P4 in the state of FIGS. 17 and 18, a width W.sub.7 of the non-display regions N1, N2, N3 and N4 recognized by a user is increased.

[0130] However, when, as in the current modified example, the non-display regions N12 and N32 of the non-display regions N1, N2, N3 and N4 in which the pad units P1, P2, P3 and P4 are respectively located are folded towards the inside of the flexible display panel 300-1 once again, the external driver or the connector C is disposed inside the flexible display panel 300-1 so that the width W.sub.7 of the non-display regions N1, N2, N3 and N4 recognized by the user is not changed.

[0131] Although the pad units P1, P2, P3, and P4 are located in the folded non-display regions N12 and N32 in the above embodiment, aspects of the present invention are not limited thereto. That is, various circuits and wirings described above may be disposed in the folded non-display regions N12 and N32.

[0132] FIG. 22 is a cross-sectional view of a flexible display panel-integrated touch panel 400 according to another embodiment of the present invention, and FIG. 23 is a cross-sectional view of a flexible display panel-integrated touch panel 400-B before non-display regions N1 and N3 of the flexible display panel-integrated touch panel 400 illustrated in FIG. 22 are folded, and FIG. 24 is an exploded perspective view of the flexible display panel-integrated touch panel 400-B illustrated in FIG. 23.

[0133] Hereinafter, the current embodiment will be described with respect to differences to the embodiments of FIGS. 1, 12, 17, and 21.

[0134] Referring to FIG. 22, the flexible display panel-integrated touch panel 400 according to the current embodiment of the present invention is constituted by attaching a touch panel 140 to the flexible display panel 100 illustrated in FIG. 1.

[0135] Referring to FIGS. 23 and 24, the touch panel 140 includes a touch region T and wiring regions T1, T2, T3 and T4 outside the touch region T. The touch panel 140 according to the current embodiment may use a variety of touch methods using capacitance overlay, a resistance film, and the like.

[0136] A flexible display panel 100-B before non-display regions are folded includes a display region D and non-display regions N1, N2, N3 and N4 outside the display region D.

[0137] The touch region T of the touch panel 140 and the display region D of the flexible display panel 100-B are aligned with each other so as to correspond to each other so that the touch panel 140 and the flexible display panel 100-B are combined with each other.

[0138] In the flexible display panel 100-B combined with the touch panel 140, the non-display regions N1 and N3 of the flexible display panel 100-B, which are on right and left sides of the display region D, are folded in a direction-z opposite to a direction z in which the display region D is displayed. As a result, the flexible display panel-integrated touch panel 400 illustrated in FIG. 22 is constituted.

[0139] Since the non-display regions N1 and N3 and the wiring regions T1, T2, T3 and T4 of the touch panel 140 are folded in the direction-z opposite to the direction z in which the display region D is displayed, when the user sees the flexible display panel-integrated touch panel 400 from the front (z-direction) of the display region D, the user recognizes that a width W.sub.8 of the non-display regions N1 and N3 as a border that surrounds the right and left sides of the display region D.

[0140] As described above, since the width W.sub.8 at which the non-display regions N1 and N3 are folded becomes much smaller than an original width W.sub.0 before the non-display regions N1 and N3 are folded, the user recognizes that the border of the display region D formed by the non-display regions N1 and N3 is significantly decreased.

[0141] Although the flexible display panel **100-B** illustrated in FIG. **1** is included in the flexible display panel-integrated touch panel **400** shown in the drawings described above, the present invention is not limited thereto, and different types of flexible display panels as those illustrated in FIGS. **1**, **12**, and **17** may be applied to the current embodiment.

[0142] FIGS. **25** through **27** illustrate examples in which the flexible display panel **100** is combined with a variety of support units. For convenience of explanation, the flexible display panel **100** is exemplified. However, the present invention is not limited thereto and may be applied to the flexible display panel-integrated touch panel **400** illustrated in FIG. **21**.

[0143] Referring to FIG. **25**, the flexible display panel **100** on which a non-display region is folded, is protected and supported by a transparent protection window **150**.

[0144] The transparent protection window **150** may be attached to the flexible display panel **100** to correspond to a shape of the flexible display panel **100**, i.e., to correspond to a top surface of the flexible display panel **100** on which an image is displayed and a folded shape of the flexible display panel **100**. That is, the transparent protection window **150** may be combined with the flexible display panel **100** so as to correspond to a shape in which the display region D and non-display regions N of the flexible display panel **100** are folded. The transparent protection window **150** has transmittance that allows light to transmit through a display screen of the flexible display panel **100** and has a strength that prevents deformation caused by a restorative force of the flexible display panel **100**. Thus, the transparent protection window **150** may protect the flexible display panel **100** from external shock or scratches and may act as a support unit that supports the shape of the flexible display panel **100**. Although not shown, an adhesive may be injected between the flexible display panel **100** and the transparent protection window **150**.

[0145] FIG. **25** illustrates a case where the transparent protection window **150** is directly disposed on the flexible display panel **100**. However, the present invention is not limited thereto and may also be applied to the flexible display panel-integrated touch panel **400** illustrated in FIG. **22**. In this regard, the transparent protection window **150** may be combined with the flexible display panel **100** to correspond to the flexible display panel **100** and/or a folded shape of the flexible display panel **100**.

[0146] Referring to FIG. **26**, an inside of the flexible display panel **100** in which the non-display regions N are folded is supported by a support member **160**. The support member **160** has a strength that prevents deformation caused by a restorative force of the flexible display panel **100**. Although not shown, the adhesive may be further injected between the flexible display panel **100** and the support member **160**.

[0147] Referring to FIG. **27**, the flexible display panel **100** of which the non-display regions are folded is supported by a flat (not curved) transparent protection window **170**. The folded portion of the flexible display panel **100** may be supported by a bezel or frame **180** that supports the flexible display panel **100** laterally. Obviously, although not shown, the adhesive may be further injected between the flexible display panel **100**, the transparent protection window **170**, and the bezel or frame **180**.

[0148] The support units described above are examples for supporting the shape of the flexible display panel including folded non-display regions, and various applications may be possible by one of ordinary skill in the art.

[0149] As described above, the flexible display panel having a folded shape supported by various support units may be integrated with various display apparatuses, such as TVs, computer monitors, PDAs, smart phones, and the like. The ratio of non-display regions to a display region of the flexible display panel recognized by a user is significantly decreased so that an enlarged image screen compared to other display apparatuses may be provided.

[0150] A display apparatus according to the embodiments described above may provide the following effects.

[0151] First, non-display regions are bent in a direction opposite to a direction in which a display

region is displayed so that a border of the display region formed by the non-display regions recognized by the user is significantly decreased.

[0152] Second, protective films are located in upper and lower portions of a flexible display panel so as to cover all of regions in which metal layers are formed, so that cracks of wirings caused by bending stress when the non-display regions are bent, may be prevented.

[0153] While this invention has been particularly shown and described with reference to preferred embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the invention as defined by the appended claims. The embodiments should be considered in descriptive sense only and not for purposes of limitation. Therefore, the scope of the invention is defined not by the detailed description of the invention but by the appended claims, and all differences within the scope will be construed as being included in the present invention.

Claims

1. A display apparatus comprising: a flexible substrate; a display region comprising a light-emitting device and disposed on the flexible substrate; a first non-display region bent with respect to the display region; a second non-display region bent with respect to the display region, the second non-display region facing the first non-display region; a third non-display region comprising a first portion being bent with respect to the display region and a second portion being bent with respect to the first portion, the third non-display region being disposed between the first non-display region and the second non-display region; a fourth non-display region disposed on an opposite side of the display region from the third non-display region, an area of the third non-display region being larger than an area of the fourth non-display region; a flexible encapsulation member disposed on the display region and the first to fourth non-display regions; a touch region disposed on the flexible encapsulation member; and touch wirings disposed outside the touch region and on the first non-display region, the second non-display region, and the third non-display region.
